A Survey on Mode of Block Cipher in Image Encryption

Vedant Kanabar
vkanabar@uoguelph.ca
1182949

Nitin Satishbabu
nsatishb@uoguelph.ca
1152910

ABSTRACT

This paper addresses the growing threat of cyberattacks and the importance of securing sensitive images while transmission. The focus of this study is determining the most effective block cipher encryption mode for images. Various methods to evaluate and contrast the cipher modes will be employed. The five block cipher modes that we will be analyzing are Electronic Code Book (ECB), CipherBlock Chaining (CBC), Cipher Feedback (CFB), Output Feedback (OFB), and Counter (CTR). The encryption and decryption of grayscale images of varying dimensions are used for this study. The research focus on the security and feasibility of implementing each mode of block cipher for image encryption. In the results, it is seen that CTR is the most feasible mode to implement in terms of security and time while CBC on average is the most secure mode of encryption.

ACM Reference Format:
Vedant Kanabar and Nitin Satishbabu. 2024. A Survey on Mode of Block Cipher in Image Encryption. *, CIS*4520 Intro to Cryptography, Final Report, University of Guelph (CIS*4520 Intro to Cryptography, Final Report, University of Guelph),* 9 pages.

1 INTRODUCTION

The surge in cyberattacks has made it imperative to safeguard sensitive image during digital transmission. Artificial Intelligence(AI) and other forms of altering images are prevalent in the society leading to fake news and misinformation. The increasing number of cases involving celebrity or influential individuals images are altered, whether for creating a negative impact or for personal benefits is a concern in today's society. This is what motivated us to study this topic of image encryption.

This paper aims to figure out the optimal block cipher encryption mode for images and also compare the modes to see if there is a significant difference in the modes. The reason we choose to study block cipher image encryption is because of the amount of resources available and studies already present in block cipher making them a feasible choice to encrypt images. Currently, there does not exist a lot of research on block ciphers and image encryption configuration optimization in terms of the configuration of the mode. The choice of mode of block cipher is very important as it had exponential differences in terms of security, time feasibility and resources needs to implement when working with large data/image sets. In today's world where people have thousands of pictures in their social media, even the smallest of advantages can be exponential when encrypting large image sets of such kind.

*CIS*4520 Intro to Cryptography, Final Report, University of Guelph, 2024*
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ACM Conference Format.

The core objective is to examine the distinctions between the original, encrypted and decrypted images then study the strengths of each mode of block cipher in order to enhance image security and contribute to ensuring the authenticity and integrity of images online. We intent to use a dataset that has grayscale images of different sizes and compositions to determine the most feasible block cipher mode for different sizes of image data sets. This research contributes in giving essential insights to enhance image security.

For this study, we will be using widely recognized block cipher algorithm such as: Advanced Encryption Standard (AES), Data Encryption Standard (DES), Triple DES (DES3), Blowfish, and ARC2. Each of these cipher algorithm will be evaluated across five different modes of operation which are: Electronic Code Book (ECB), Cipher Block Chaining (CBC), Cipher Feedback (CFB), Output Feedback (OFB), and Counter (CTR). By analyzing the performance of these cipher modes in the context of images, we aim to improve image security in today's digital world.

2 RESEARCH QUESTIONS

(1) Which block cipher mode has the most security in image encryption?
(2) Which block cipher mode is the most feasible to implement for big image data set encryption?

3 BACKGROUND OR LITERATURE REVIEW

From researches in Block Cipher Modes of Operation in plaintext, the results that "The Counter (CTR) mode is the most secure, efficient, and fastest way of doing encryption" while "The Cipher Block Chaining (CBC) mode can be considered as a secure encryption scheme". ECB on the other hand "Should not be practiced for general purposes since it lacks essential security requirements"[4].

In terms of encryption algorithms AES can be effective in comparison to the other mode of operations for images[6]. AES is considered fast for encrypting images when compared to other encryption algorithms and can be used be general computer/devices[15]. We will be evaluating the most secure and most feasible block cipher mode for images. All other algorithms used in this paper (Blowfish, DES, DES3, RC2) are for research purposes and should not to used as they have been deprecated.

3.1 Block Cipher Mode of Operation

Block cipher is a method for encrypting plaintext into blocks to produce ciphertext using a key and algorithm. In this method, each block of plaintext is treated as a whole and transformed into the ciphertext of similar length (typically, 64 or 128 bits).

Mode of Operation is divided into two consisting of Deterministic Encryption and Probabilistic Encryption. The Electronic Code Book (ECB) falls under Deterministic Encryption. Probabilistic Encryption further divides into two consisting of Block Cipher and Stream Cipher. Cipher Block Chaining (CBC) falls under Block

cipher. While Stream Cipher encompasses the following: Cipher Feedback (CFB), Output Feedback (OFB) and Counter (CTR)[14]. This will be further discussed in the following sections.

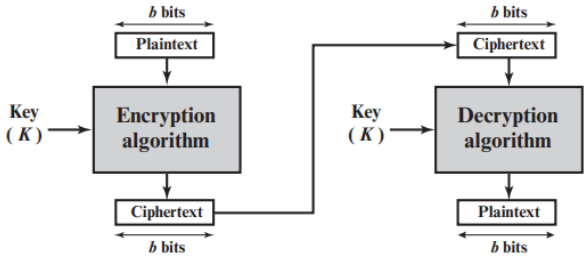


Figure 1: Block Cipher [12]

3.2 Electronic Code Book (ECB)

In the Electronic Code Book (ECB) mode of operation, each block of plaintext is encrypted separately using the same key, which results in independent cipher blocks. The decryption involves using the encrypted block of cipher text and applying the decryption algorithm. This can be seen in Figure 2. This mode requires only a key and does not require complex chaining or initialization vectors.

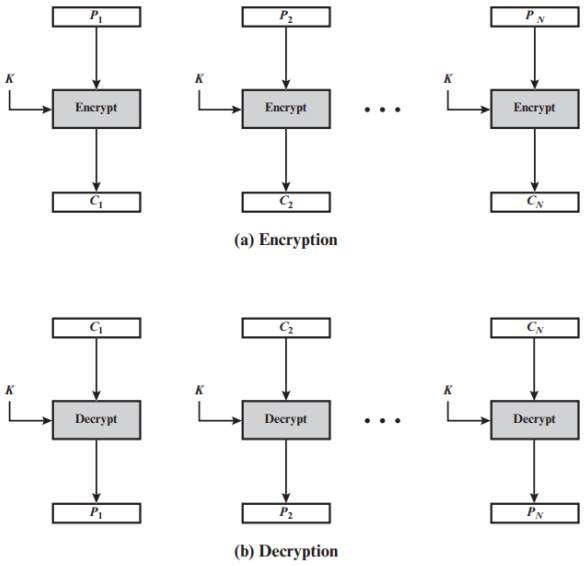


Figure 2: ECB Mode Encryption and Decryption Process [12]

The main advantages of ECB are that it is simple and efficient for encrypting blocks of plaintext. Since, there is no dependencies or chaining, this mode is suitable for parallel processing. Furthermore, there is no error propagation since the errors are contained in a particular block and do not propagate to the remaining blocks during decryption. There is a lack of diffusion meaning it is not recommended for large amount of plaintext or plaintext with patterns, as it does not provide security.

3.3 Cipher Block Chaining (CBC)

In the Cipher Block Chaining (CBC) mode of operation, each plaintext block is XORed with the previous ciphertext block or initializing vector (IV) for the first block before encryption using the key. For decryption each ciphertext block is decrypted using the key and XORed with the previous ciphertext block to attain the original plaintext block. This mode requires a key and initializing vector (IV). As depicted in the Figure 3.

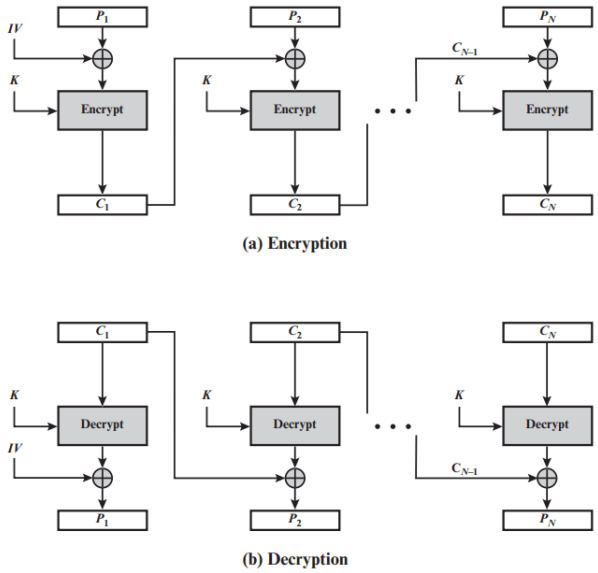


Figure 3: CBC Mode Encryption and Decryption Process [12]

The main advantages of CBC are that it provides diffusion and enhances secutiry. By XORing each plaintext with the previous ciphertext block before encryption it makes it suitable for encrypting plaintext with patterns. Encryption cannot be parallelized and results in slower encryption time. However, decryption can be parallelized.

3.4 Cipher Feedback (CFB)

In the Cipher Feedback (CFB) mode of operation, a block cipher is transformed into a stream cipher. This mode functions on segments of s bits. s usually set to 8. Encryption starts by initializing the b-bit shift register with an initialization vector (IV). The leftmost s bits of the encryption function output are XORed with the plaintext producing the ciphertext. In decryption, the ciphertext is XORed with the output of the encryption function to produce the plaintext. As depicted in the Figure 4.

The main advantages of CFB are that only encryption function is utilized for both the encryption and decryption process and the plaintext does not need to be padded to be a multiple of the cipher block size. The disadvantages are that the errors/changes in the plaintext does propagate to the other blocks (though limited) and encryption cannot be parallelized and results in slower encryption time. However, decryption can be parallelized.

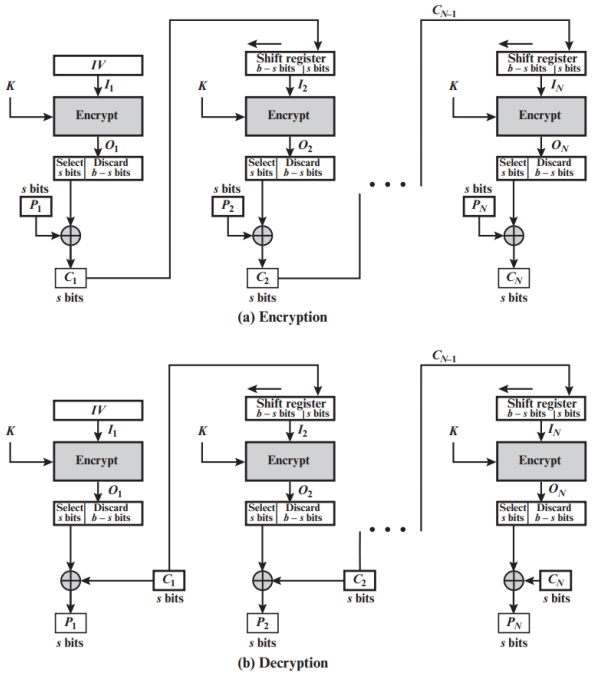


Figure 4: CFB Mode Encryption and Decryption Process [12]

3.5 Output Feedback (OFB)

In the Output Feedback (OFB) mode of operation, a block cipher is transformed into a stream cipher. The output of encryption function is used as input for the next block of plaintext and also XORed with plaintext to produce the ciphertext. Plaintext and ciphertext functions as full blocks.

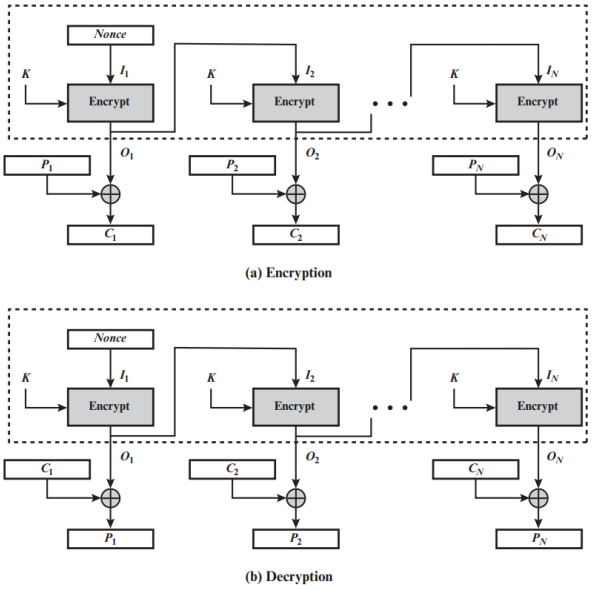


Figure 5: OFB Mode Encryption and Decryption Process [12]

The main advantages in OFB are that only encryption function is utilized for both the encryption and decryption process and bit errors in transmission do not propagate. The disadvantage is that both encryption and decryption cannot be parallelized this may impact total encryption and decryption times.

3.6 Counter (CTR)

In the Counter (CTR) mode of operation, turns a block cipher into a stream cipher by generating a unique keystream for each block using a counter. The nonce, counter and key are inputs for the encryption function, the resulting output is then XORed with the plaintext to produce the ciphertext [2]. In decryption, similar to encryption, the nonce, counter and key are the inputs for the encryption function, the result is XORed with the ciphertext to produce the plaintext. As depicted in the Figure 6.

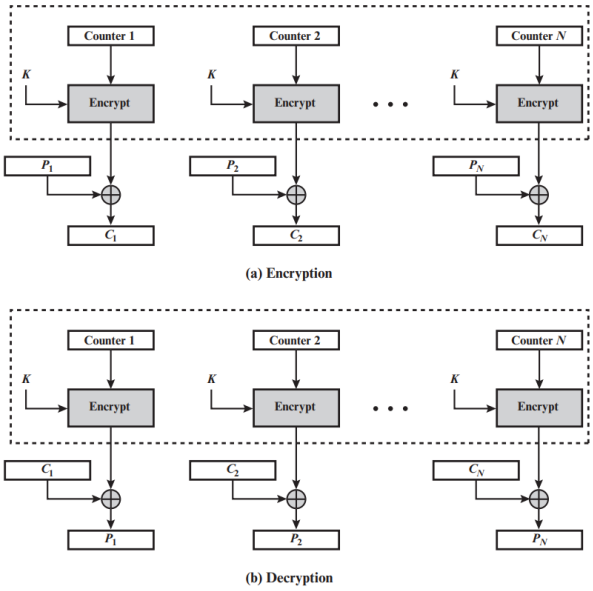


Figure 6: CTR Mode Encryption and Decryption Process [12]

The main advantages in CTR are that only encryption function is utilized for both the encryption and decryption process and that the encryption and decryption process can be parallelized resulting in decreased output times. One disadvantage is that the nonce needs to be unique for each message.

4 EXPERIMENTAL SETUP

In this section, we will discuss the experimental setup and the hardware and software requirements as well as the dataset selection and set up.

4.1 Experimental Steps

To carry out the experiment, the full process flow is show in Figure 7. First the image from the dataset is converted into gray scale and strored. Then Image analysis based on performance metrics (See Section 5) is carried out on the original gray scale image. The image

is then encrypted on choice of cipher and mode of cipher. Image analysis is carried out on the encrypted image. Then the image is decrypted and image analysis is once again carried out on the decrypted image.

All image analysis performance metrics are discussed in Section 5. Further information about hardware, software and dataset is below.

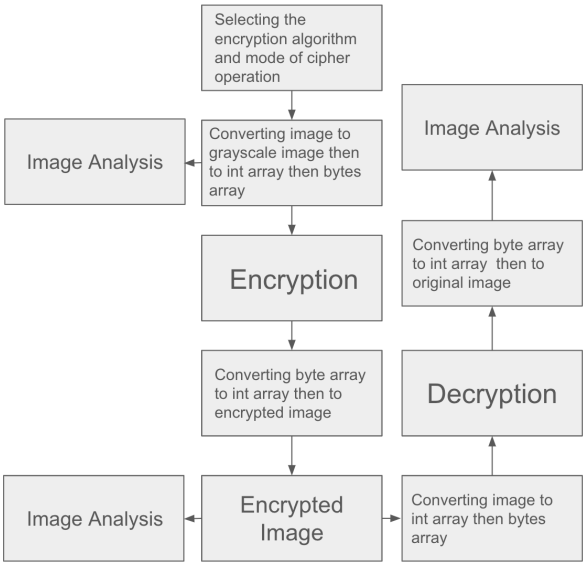


Figure 7: Image showing experiment process and work flow

4.2 Hardware and Software setup

Below in Table 1, the hardware and software components are described as well as the algorithm properties for each of the five algorithms applied (AES, Blowfish, DES, DES3, RC2).

4.3 Dataset Selection

- 2 JPG Images of size 256x256 [10]
- 2 JPG Images of size 512x512 [11]
- 1 JPG Image of size 2000x2000 [13]

All images were converted to gray scale. This choice was made as in a colour mode, the same process will be carried out three times for R,G,&B (0-255) representation of the image [8] and (0-255) narrows down the options for the algorithms to choose pixels from making it harder to encrypt. A good encryption in such harder conditions ensures our results are more applicable for general purpose.

These Dataset contains images ranging from the size used in a profile picture to a images that can be captured by professional cameras. Securing these images are vital when it comes to protecting user identity and confidentiality. A big sized 2000x2000 was chosen to analyze encryption and decryption times.

5 PERFORMANCE METRICS

In order to have a thorough understanding of which mode is the best block cipher mode of operation a wide variety of performance parameters/metrics have been measured for each of the given encryption modes and algorithms. In this section, we will discuss all these performance metrics measured in the experiment.

	Specifications
Processor	13th Gen Intel(R) Core(TM) i7-13700F 2.10 GHz
Memory	RAM 32.0 GB (31.8 GB usable)
System type	64-bit operating system, x64-based processor
Operating System	Windows
Python	v3.12.2 (to program the encryption and decryption scripts)
OpenCV	v4.9.0 (an Open Source Computer Vision Python library to help with manipulating and extracting image properties)
Numpy	v1.26.3 (a Python library for scientific computing in Python)
Mathplotlib	v3.8.4 (a Python library for creating static, animated, and interactive visualizations)
Crypto.Cipher	v1.4.1 (a Python package that contains algorithms for protecting the confidentiality of data)
Scipy.stats	v1.13.0 (a Python package to calculate pearson correlation and other statistics)
Visual Studio Code	our choice of code editor (any code editor can be used)
AES	Key - 32 Bytes, IV - 16 bytes, Nonce - 8 bytes
Blowfish	Key - 32 Bytes, IV - 8 bytes, Nonce - 4 bytes
DES	Key - 8 Bytes, IV - 8 bytes, Nonce - 4 bytes
DES3	Key - 16 Bytes, IV - 8 bytes, Nonce - 4 bytes
RC2	Key - 32 Bytes, IV - 8 bytes, Nonce - 4 bytes

Table 1: Setup Parameters

5.1 Encryption/Decryption Time

In order to answer our RQ2 of feasibility to encrypt a large set of images, the encryption and decryption time make a huge difference. To evaluate this, we will run each mode with a large image size of 2000x2000 pixels to simulate a large image set although in real life a large image set would be a lot greater. Hence a small difference in encryption/decryption time in the 2000x2000 image will make a huge butterfly effect to larger image sets.

5.2 Visual Assessment

A good encryption techniques is one that does not leave behind any clues or hints about what the original image is and instead

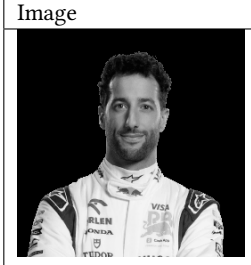
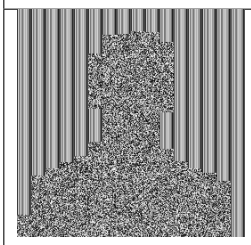
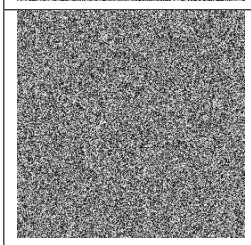
Image	Comments
	Original
	Can make out original
	Can see outline
	Cannot make out original image

Table 2: Categories of Visual Assessments of Images

produces a fully different image. The encrypted image should be arbitrary and randomly arranged such that no intruder can make out what the original image is. In order to get a good assessment of the Visual image, each encrypted image was put into one of three categories; Can make out original, Can see outline and Cannot make out original image. These categories are further described in the Table 2.

5.3 Histogram analysis

In order to see the spread of pixel values, we can use a histogram to see a graphical analysis of the frequency distribution of the pixel intensities (0-255) in gray scale mode. In the cipher image, it is ideal that this frequencies have a perfectly uniform distribution as this implies the encryption is secure and there is no leak of information like frequency of pixel intensities to attackers.

In order to evaluate this comparison between a perfectly uniform distribution, a chi test against a perfectly uniform distribution is carried out using the following Equation 1:

Images	Correlation coefficient		
	H	V	D
Input Image	0.9492	0.9801	0.9398
Cipher Image	0.0019	0.0079	0.0046
Decrypted Image	0.9492	0.9801	0.9398

Table 3: Comparison results of the correlation coefficient

$$\chi^2 = \sum_{p=0}^{255} \frac{(E_p - O_p)^2}{O_p} \tag{1}$$

Where m, n are the width and length of image in pixels, E_p is the expected frequency of a given pixel intensity (0-255) calculated by $E_p = \frac{mn}{256}$ and O_p is the observed frequency of a given pixel intensity (0-255). At significance level $\alpha = 1\%$ and $\alpha = 5\%$ with 255 degrees of freedom, the critical chi-square values to pass the hypothesis uniformity are $\chi^2(0.01, 255) = 310.458$ and $\chi^2(0.05, 255) = 293.248$, respectively. A good encryption means the chi test score value will be below both these critical values.

5.4 Adjacent Pixel Correlation analysis

In a normal image, there exists a certain correlation/relationship between pixels and their adjacent pixels. Due to recurrent pixels in normal images, there exits strong correlations between adjacent pixels with a dense correlation graph. A cipher image on the other hand should always have low or ideally no correlation between adjacent pixels be it Horizontal, Vertical or Diagonal. The equation to calculate the correlation coefficient is below Equation 2:

$$r_{(x,y)} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{2}$$

where,

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i \tag{3}$$

and x_1 is the pixel intensity value and y_1 is the corresponding horizontal, vertical or diagonal adjacent pixel intensity value.

For Horizontal correlation, the pair are made by pixel value at (x, y) and pixel value at $(x, y+1)$, in vertical pixel value at (x, y) and pixel value at $(x+1, y)$ and in diagonal pixel value at (x, y) and pixel value at $(x+1, y+1)$.

From Figure 8, it can be seen that the graph of cipher images is uniformly distributed and randomly spread out. The correlation values corresponding to the graphs in Figure 8 are shown in Table 3. The ideal value of the correlation coefficient is 0 which we can see is closely achieved in the given Cipher image [5].

5.5 Shannon Entropy

Claude Shannon gave the measurement of the amount of information contained in data and he termed it as entropy [3]. In a gray-scale image, we have pixel intensities from (0-255). In this case, the Shannon entropy is calculated using the following Equation 4:

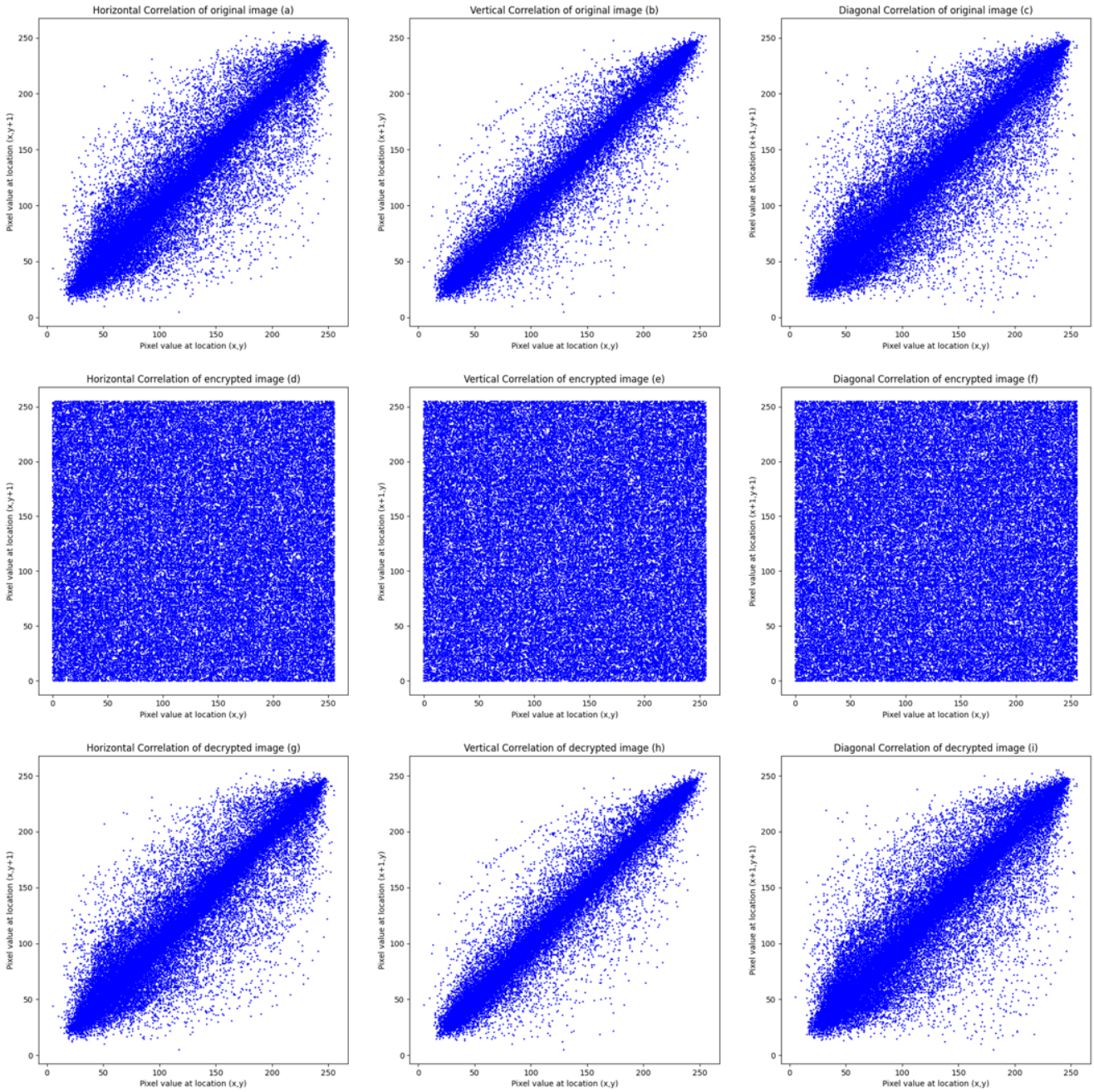


Figure 8: Image showing Correlation analysis of gray scale images: (a-c) shows the correlation of the input man.jpg image, (d-f) shows the correlation of the cipher of man.jpg, (g-i) shows the correlation of the decrypted image of man.jpg

$$H = \sum_{i=0}^{255} P_i \log_2(P_i) \tag{4}$$

Where P_i is the probability of a pixel of having an intensity i. Ideally we want a uniform distribution of intensities which in our case leads to a Shannon entropy close to 8. The closer we are to 8, the more secure the images is.

6 RESULTS

This section discusses the results summarized for each performance parameter.

6.1 Encryption/Decryption Time

On average looking at the encryption and decryption times on Table 4, it is clear that CTR and ECB were had one of the fastest encryption times with on average either the lowest encryption time or lowest ratio compared to the lowest encryption time. This is expected behaviour as CTR and ECB have the least amount of calculations while encrypting and each block can also be encrypted in parallel. Although not implemented in this experiment, the parallel implementation of CTR and ECB imply a lot faster encryption and decryption time for very large sets of data in comparison to other modes. In this aspect, these two modes are superior to the other modes of operation.

6.2 Visual Assessments

As depicted from the example in Table 5, all encrypted images of the different encryption algorithm and block cipher modes were visually indistinguishable. They did not show traces of original images and did not resemble specific aspects of the original image.

6.3 Histogram Analysis

Referring to Table 6, the frequency of the pixel values did show a uniform distribution for almost all encrypted images except Blowfish-ECB, DES-ECB, DES3-ECB and RC2-ECB. The critical chi-square values to pass the hypothesis uniformity are $\chi^2(0.01, 255) = 310.458$ and $\chi^2(0.05, 255) = 293.248$, respectively. The chi scores fall below the critical values for uniformly distributed encrypted images, ranging from 230.997 (AES-CTR) to 273.185 (AES-ECB). This indicates that the encryption process is secure. While, outliers in chi score, such as Blowfish-ECB, DES-ECB, DES3-ECB and RC2-ECB are not uniformly distributed in terms of pixel values and may not be secure for transmission.[9] Overall this indicates how much weaker ECB block cipher mode is in terms of security compared to other modes

6.4 Adjacent pixel Correlation Analysis

Looking at Table 7, we can see in general, most encryption modes and techniques have a adjacent pixel correlation of <0.01 with a few outliers in ECB mode once again highlighting its weaknesses. Again referring back to Figure 8, we can see in general how much more spread out the data points of adjacent pixels are in comparison to the original images. Strong and secure cipher images should have a low correlation among the adjacent pixels with an evenly distributed graph. [7] In general that is depicted in all modes but relatively weakest in ECB mode. The encryption process with lowest average correlation were: AES-CTR, Blowfish-CFB, DES-CFB, DEC3-OFC, RC2-CFB

6.5 Shannon Entropy

The Table 8 depicts the average Shannon entropy values of encrypted images using various algorithms and modes. As previously stated, these Shannon entropy values are close to 8 indicating that there is high level of randomness resulting in improved security [1]. Overall all the displayed high Shannon entropy values, with the lowest being 7.9971428347 (DES-ECB). The encryption process that showed the highest Shannon entropy values were: AES-CTR, Blowfish-CBC, DES3-CFB, RC2-CFB and DES-OFC.

7 CONCLUSION

From the accumulation of the various evaluation methods, we have assigned a score for each cipher mode in table 9. We can see that CBC mode excels when it comes to the security of images. Similar to plaintext [4] CBC mode performs exceptionally in the case of images as well. CTR, OFB, CFB perform are closely behind the CBC mode in term of security. Each excelling in a particular field when compared amongst themselves the results show that: CTR performs best when it comes encryption/decryption time, OFB performs best in adjacent pixel correlation and CFB in Shannon entropy randomness. Lastly, ECB is not recommended for image encryption since it lacks security even though it performs well on encryption/decryption times.

Feasibility to implement is a major factor when considering an encryption process. When considering big image dataset CTR stands out significantly, further the encryption and decryption can be parallelized decrease the time required. Overall CTR is a solid option when taking into account feasibility and security. CBC and OFB are moderately feasible for large image datasets with average encryption/decryption times similar to one another. ECB produces the lowest encryption/decryption times, but must be avoided due to the lack of security. CFB having the longest average time for encryption/decryption is not feasible for the public devices although secure. All these evaluations are presented in the Figure 9

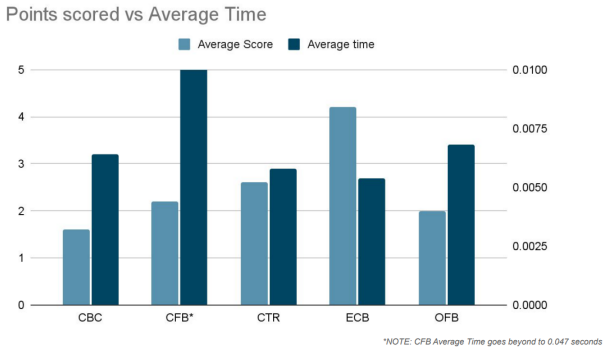


Figure 9: Points Scored Vs Average Time

8 FUTURE WORKS

In terms of Future research, on can build on this to choose a better image block zoning method like Zig-Zag transform and DNA transform[16] to further improve security in all aspects. When implementing a new block cipher algorithm specifically for images, this research information can help provide guidance on optimization for mode of operation. In general, these results can be used to help encrypt images on a large scale which is significant in today's digital age where people have the ability to capture and store thousands of pictures on their phones and social media platforms that contain millions of pictures everyday.

REFERENCES

[1] José A.P. Artiles, Daniel P.B. Chaves, and Cecilio Pimentel. 2019. Image encryption using block cipher and chaotic sequences. *Signal Processing: Image Communication* 79 (2019), 24–31. https://doi.org/10.1016/j.image.2019.08.014

Algorithm	CBC		CFB		CTR		ECB		OFC	
	Average Enc/Dec Time (s)	Ratio of Lowest time	Average Enc/Dec Time (s)	Ratio of Lowest time	Average Enc/Dec Time (s)	Ratio of Lowest time	Average Enc/Dec Time (s)	Ratio of Lowest time	Average Enc/Dec Time (s)	Ratio of Lowest time
AES	0.0005	1.0000	0.0072	14.4870	0.0005	1.0000	0.0005	1.0000	0.0018	3.5450
Blowfish	0.0028	2.7469	0.0146	14.5896	0.0010	1.0000	0.0010	1.0384	0.0034	3.3545
DES	0.0053	1.4947	0.0299	8.5155	0.0035	1.0000	0.0036	1.0336	0.0040	1.1349
DES3	0.0118	1.0488	0.0916	8.1326	0.0115	1.0212	0.0113	1.0000	0.0121	1.0704
RC2	0.0058	1.0378	0.0557	9.9781	0.0072	1.2842	0.0056	1.0000	0.0079	1.4087

Table 4: Comparison results of the encryption/decryption time

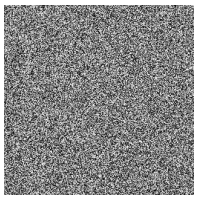
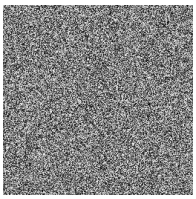
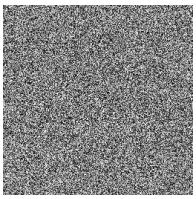
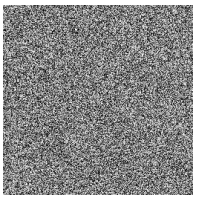
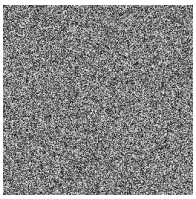
Original Image	Encrypted Images in AES algorithim				
	CBC	CFB	CTR	ECB	OFC
					

Table 5: Comparison results of the visual assessment

Algorithm	Average Chi-Score of Encrypted Images				
	CBC	CFB	CTR	ECB	OFC
AES	251.309	248.173	230.997	273.185	270.991
Blowfish	252.491	251.304	253.768	443.827	264.255
DES	262.565	260.171	253.564	462.809	252.887
DES3	247.804	243.807	251.126	460.012	253.218
RC2	255.936	237.687	242.286	438.595	267.659

Table 6: Comparison results of the Histogram chi score

Algorithm	Average H,V,D Correlation Coefficient of Encrypted Images				
	CBC	CFB	CTR	ECB	OFC
AES	0.000865	0.000680	0.000238	0.004300	-0.001196
Blowfish	0.000279	0.000238	-0.000397	0.012158	0.000887
DES	-0.000541	-0.000075	-0.000251	0.010135	-0.000279
DES3	0.000287	0.000245	0.002000	0.009320	-0.000180
RC2	-0.000290	-0.000233	0.002063	0.009533	0.001361

Table 7: Comparison results of the correlation coefficients

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Algorithm	Average Shannon Entropy of Encrypted Images				
	CBC	CFB	CTR	ECB	OFC
AES	7.9985140661	7.9986262558	7.9987024642	7.9982592726	7.9985003456
Blowfish	7.9985540266	7.9985237550	7.9985006962	7.9973069620	7.9984644072
DES	7.9984308346	7.9985033070	7.9985117142	7.9971428347	7.9985284614
DES3	7.9985849114	7.9986815552	7.9985023490	7.9973316481	7.9984638794
RC2	7.9984883769	7.9986036836	7.9985440266	7.9974207267	7.9984763950

Table 8: Comparison results of the Shannon Entropy

Mode of Operation	Visual Assessment	Encryption/ Decryption Time (s)	Histogram	Correlation	Shannon Entropy	Average Ranking
CBC	Passed	0.006403	H_0 was Accepted at $\alpha = 0.05$	0.00011986	7.998514443	1.8
CFB	Passed	0.047963	H_0 was Accepted at $\alpha = 0.05$	0.00017118	7.998587711	2.2
CTR	Passed	0.005797	H_0 was Accepted at $\alpha = 0.05$	0.00073047	7.998552250	2.2
ECB	Passed	0.005380	H_0 was Declined at $\alpha = 0.01$	0.00908926	7.997492289	4.2
OFB	Passed	0.006818	H_0 was Accepted at $\alpha = 0.05$	0.00011873	7.998486698	2.2

Table 9: Overall comparison of results and assessments

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